

Transition waves in a beam coupled to a foundation of symmetric bistable elements

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Multistable metamaterials, capable of adopting multiple stable configurations, offer versatile control over the shape and mechanical properties of systems. Transition waves (TWs), which propagate spatially through state transitions, provide a promising mechanism for reconfiguring such materials. While TWs in systems with asymmetric energy landscapes propagate stably by transitioning from higher to lower energy states, they require additional energy input to reset the structure, limiting their reusability. This work explores a multistable system with a reconfigurable surface shape, comprising an elastic slender beam coupled to a foundation of bistable elements with a symmetric energy landscape. The symmetric landscape enables TWs with tunable speeds, energy, and propagation distances, which can be controlled by the impact speed on the boundaries. The stop position of the TW forms a static domain wall, separating regions into distinct stable states. Leveraging the relationship between impact speed and propagation distance, we propose a dynamic strategy for reversible surface shape reconfiguration using sequential impulses to generate targeted configurations. An experimental demonstration using buckled double beams as bistable elements confirms the feasibility of this strategy, highlighting its potential for postmanufacturing shape control in multistable systems.

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I. INTRODUCTION

Multistable mechanical metamaterials, usually comprising periodically arranged multistable elements that enable multiple global stable configurations, have been used for shape reconfiguration [1,2], reusable energy absorption [3], and tunable mechanical properties and functionalities [4–7]. In these systems, when a specific multistable element is changed from one stable state to another, it is possible for the state (phase) to spatially propagate to adjacent multistable elements in the form of transition waves (TWs). The conditions for TW propagation have been investigated theoretically [8–12], numerically, and experimentally [13–22]. For systems composed of identical units with asymmetric energy landscapes, TWs can propagate stably with constant speed and constant wavefront shape. The local transition from a higher energy state to a lower energy state releases sufficient energy to continue the propagation despite dissipation in the system. This allows a TW to be used to efficiently reconfigure an entire system despite being initiated by a single localized load. Examples

include a chain of bistable inclined beams coupled by elastic elements [13], a chain of bistable plates coupled by magnetic fields [14,15], a system of rotating squares with embedded magnets [16,21,22], a one-dimensional (1D) linkage [17], a two-dimensional (2D) multistable kirigami structure [18], a 2D network of hexagonal blocks [19,20], and hierarchical multistable metamaterials [23].

However, asymmetric systems also bring intrinsic limitations for complex reconfiguration. One limitation is that the TWs propagate only for transitions from the high energy state to the low energy state. Consequently, the reconfiguration enabled by TWs is one-way and single-use, requiring additional energy input to reset the system to the higher energy state before another TW can propagate. Moreover, TWs in asymmetric systems tend to propagate throughout the entire structure, limiting the complexity of the reconfiguration pattern. If more complex target patterns are desired (requiring control of the motion of TWs, such as the propagation path and stop position), heterogeneities or defects are necessary [18,24]. Even with such modifications, the reconfiguration is often limited to a single target [18], requiring either the construction of a new mechanical system for each desired configuration or the

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incorporation of additional control mechanisms, such as electrical or magnetic approaches for real-time modulation of the bistable potential [25,26], which increases the overall system complexity. In contrast, arrays composed of bistable elements with two equal energy minima (i.e., “symmetric systems”) [6,27,28] allow for two-way TWs (dynamic kinks and antikinks, which are used interchangeably with TWs throughout this paper), enabling transitions between either of the two states and thus overcoming the unidirectional nature and the need to inject more energy into the system to reset it after a TW has propagated. Additionally, in symmetric arrays, the TW propagation distance is limited by dissipation and is directly influenced by the input energy [28]. The intrinsic dissipation in any real physical system can be leveraged to stop TWs at specific locations without relying on defects. Once a kink (or antikink) stops, it forms a static domain wall (DW) in the metamaterial. The spatially periodic energy barriers and local minima arising from the system’s discreteness (the Peierls-Nabarro effect) can then pin and stabilize the static DW against small perturbations [29]. These properties of symmetric systems enable a strategy for reconfiguring mechanical metamaterials among multiple stable configurations by generating sequential dynamic kinks and antikinks with controlled energy and propagation distances through boundary impacts, which have not been explored in previous research.

Previously, multistable systems have primarily consisted of open cells that reconfigure between discrete geometries [13–28]. However, coupling such multistable behaviors to surfaces would enable the proposed dynamic reconfiguration of surface morphing that can be relevant in a variety of applications, including morphing airfoils [30], adaptive optical devices [31], and flow control [32,33]. In this work, we design a multistable system with a reconfigurable surface shape, comprising an elastic slender beam coupled to a foundation of bistable elements with a symmetric energy landscape [Figs. 1(a) and 1(b)]. The deformation of the slender beam provides the continuous surface shape while the symmetric bistable foundation enables the proposed postmanufacture reconfiguration strategy. We characterize the behavior of the TWs of this system theoretically, numerically, and experimentally. Through theoretical analysis and numerical simulations, we show that the system supports two-way TWs [kinks and antikinks, Fig. 1(a)] associated with the transitions between the two energy minima in the symmetric energy landscape. Moreover, the speed and energy of the TWs in this system can be controlled via the speed of the impact used to initiate the TWs. Due to dissipation, TWs decelerate, associated with a loss of kinetic energy. Eventually, they stop at a specific location in the form of a static DW, separating regions of the system into different stable states and forming an undulating surface shape. Based on the quantitative relationship between boundary impact

speed and wave propagation distance, we demonstrate a dynamic strategy to reconfigure the surface shape simply by applying multiple impulses on the boundaries of the system. Targeted configurations can be achieved via a sequence of multiple TWs with controlled initial speeds (and, hence, controlled propagation distance). The above is experimentally validated using a foundation of symmetric bistable elements comprising a buckled double-beam mechanism, demonstrating the feasibility of this dynamic reconfiguration strategy.

II. THEORETICAL MODEL AND NUMERICAL METHOD

The slender beam is connected to the ground via a bistable foundation comprising distributed bistable elastic elements [Fig. 1(a)]. Each bistable element has two stable equilibrium positions, referred to as state O (original) and state B (compressed) [Fig. 1(b)], with identical elastic potential energies, $U_b(w_O) = U_b(w_B)$. These elements provide a distributed vertical force acting on the beam. Within this system, the snap-through behavior of individual elements, as they transition from one stable state to the other, can propagate from an initiation point to the surrounding regions. To study the properties of the associated TWs, we first derive the governing equation of the system dynamics. To simplify the model, three assumptions are made. (1) The beam’s aspect ratio and deflection are small, allowing us to neglect its elongation and shear deformation. Consequently, the beam’s deformation is fully described by its deflection, $w(x, t)$, following the Euler-Bernoulli beam theory (a comparison with the Timoshenko beam theory is in the Supplemental Material [34]). (2) The foundation interacts with the beam solely through a distributed vertical force, F , and the bistable force-displacement relation [Fig. 1(b)] follows the ϕ^4 model [35], where the F - w relationship is described by a cubic function. (3) The foundation density is large enough compared to the characteristic width of the TW and thus the foundation can be approximated by a homogenous continuous foundation. Under these assumptions, the beam deflection, $w(x, t)$, is governed by the following fourth-order nonlinear partial differential equation [36]:

$$EI \frac{\partial^4 w}{\partial x^4} + A\rho \frac{\partial^2 w}{\partial t^2} + \beta \frac{\partial w}{\partial t} + kw(w - 0.5w_B)(w - w_B) = 0 \quad (1)$$

where E, I, A, ρ, β, k are the Young’s modulus, moment of inertia, cross-section area, density of the beam, the linear damping, and the bistable foundation stiffness, respectively. Here we use a simple linear viscous-type damping term to represent the first-order approximation of all the possible dissipation mechanisms in a real physical system for model simplicity. To normalize the governing equation,

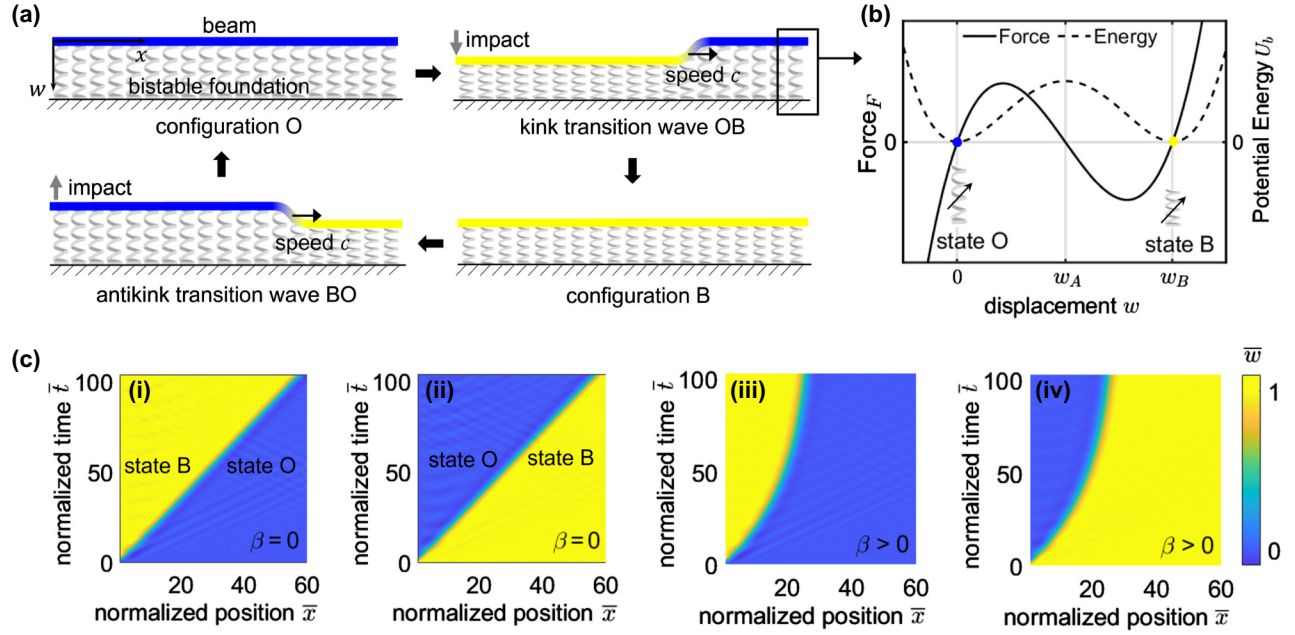


FIG. 1. (a) Schematic of the system and two-way transition waves (TWs): an elastic beam is connected to the ground by a foundation of bistable elements. (b) Elastic potential energy landscape and force-displacement curve per unit length. (c) Spatiotemporal plot when a TW propagates: (i),(ii) two-way TWs without damping $\beta = 0$ and (iii),(iv) with damping $\beta > 0$.

we define the characteristic length L , time T , and speed V as follows:

$$L = \sqrt[4]{\frac{EI}{kw_B^2}}, \quad T = \sqrt{\frac{\rho A}{kw_B^2}}, \quad V = L/T, \quad (2)$$

and then Eq. (1) can be normalized as

$$\frac{\partial^4 \bar{w}}{\partial \bar{x}^4} + \frac{\partial^2 \bar{w}}{\partial \bar{t}^2} + \bar{\beta} \frac{\partial \bar{w}}{\partial \bar{t}} + (\bar{w}^3 - 1.5\bar{w}^2 + 0.5\bar{w}) = 0 \quad (3)$$

with dimensionless variables $\bar{w} = w/w_B$, $\bar{x} = x/L$, $\bar{t} = t/T$, and $\bar{\beta} = \beta/\sqrt{\rho A k w_B^2}$.

To confirm the existence of a stable TW in the system, we assume the TW propagates at a constant speed $\bar{c}$, and introduce the traveling coordinate $\xi = \bar{x} - \bar{c}\bar{t}$ into Eq. (3):

$$\frac{d^4 \bar{w}}{d\xi^4} + \bar{c}^2 \frac{d^2 \bar{w}}{d\xi^2} - \bar{\beta} \bar{c} \frac{d\bar{w}}{d\xi} + \bar{U}'(\bar{w}) = 0 \quad (4)$$

where $\bar{U}'(\bar{w}) = \bar{w}^3 - 1.5\bar{w}^2 + 0.5\bar{w}$. Multiplying Eq. (4) by $d\bar{w}/d\xi$, integrating by parts over $\xi = (-\infty, +\infty)$ and noting that $\frac{d\bar{w}}{d\xi} = 0$ and $\frac{d^2 \bar{w}}{d\xi^2} = 0$ at the boundaries (see the Supplemental Material [34] for details), one obtains

$$\begin{aligned} \bar{\beta} \bar{c} \int_{-\infty}^{+\infty} \left(\frac{d\bar{w}}{d\xi} \right)^2 d\xi &= U(w(\xi = +\infty)) - U(w(\xi = -\infty)) \\ &= U(0) - U(w_B) = \Delta \bar{U}_{OB}. \end{aligned} \quad (5)$$

The left-hand side of Eq. (5) is always greater than or equal to zero, requiring that the right-hand side $\Delta \bar{U}_{OB} \geq 0$. This reveals the necessary condition for a stable TW: elastic potential energy released by the foundation should be equal to energy dissipation. Consequently, this condition excludes the existence of a stable TW from a low energy state to a high energy state. When $\Delta \bar{U}_{OB} > 0$ and $\bar{\beta} > 0$, a specific wave speed $\bar{c}$ can be determined from Eq. (5), which is consistent with previous research on TWs in systems with asymmetric energy landscapes [12]. However, in this work, the symmetric energy landscape shown in Fig. 1(b) leads to $\Delta \bar{U}_{OB} = 0$. As a result, both the kink TW OB [Fig. 1(c)(i)–(ii)] and the antikink TW BO are possible [Fig. 1(c)(iii)–(iv)] and share the same dynamic properties.

In an early study [37] on quasistatic and steady-state buckle propagation in a beam on a nonlinear foundation exhibiting snap-through behavior, the authors showed that stable buckle propagation occurs when the two areas enclosed by the intersection of the external load line and the force-displacement curve of the nonlinear foundation are equal. This critical external load is also referred to as the “Maxwell line” in the context of phase transformations [38]. The concept of a symmetric energy landscape in this work essentially corresponds to the equal-area condition defined by the Maxwell line. Next, we consider two damping cases in this symmetric system. First, when $\bar{\beta} = 0$, any constant wave speed $\bar{c}$ conceptually satisfies Eq. (5), implying the system supports TWs with different wave speeds. In this case, the speed is not uniquely determined by Eqs. (4) and (5) but instead depends on

the initial conditions and boundary conditions. In the second case, when $\bar{\beta} > 0$, Eq. (5) is valid only for $\bar{c} = 0$, corresponding to the static deformation solution, which could either be a global flat state or a static state with a DW. This indicates that it is impossible to maintain a constant wave speed for $\Delta\bar{U}_{OB} = 0$ and $\bar{\beta} > 0$. During propagation, the system cannot compensate for energy dissipation by releasing stored elastic energy from the foundation. Consequently, the kinetic energy diminishes, the TW decelerates, and eventually the TW halts after propagating a certain distance.

To verify the above theoretical analysis, we next numerically solve Eq. (3). The spatial domain is discretized using the second-order central difference method, while the fourth-order Runge-Kutta method is employed for time integration. The dimensionless length of the beam, $\bar{l}$, is chosen to be sufficiently large to prevent wave reflections at the boundaries. To initiate a TW, the left boundary is subjected to an impact, causing it to snap from one state to the other at a prescribed impact speed V_{imp} . For instance, to initiate a kink TW (from state O to B), the following boundary condition (BC) is used:

$$\bar{w}(\bar{t})|_{\bar{x}=0} = \begin{cases} \bar{V}_{\text{imp}}\bar{t}, & \text{if } 0 \leq \bar{t} \leq \frac{1}{\bar{V}_{\text{imp}}}, \\ 1, & \text{otherwise,} \end{cases} \quad (6)$$

where the nondimensional impact speed is normalized by the reciprocal of T of Eq. (2). The other boundary conditions are

$$\frac{\partial^2 \bar{w}}{\partial \bar{x}^2} \Big|_{\bar{x}=0} = \frac{\partial^2 \bar{w}}{\partial \bar{x}^2} \Big|_{\bar{x}=\bar{l}} = \frac{\partial^3 \bar{w}}{\partial \bar{x}^3} \Big|_{\bar{x}=\bar{l}} = 0.$$

Additionally, to validate the model described by Eq. (3) and the numerical results, a 2D beam model supported by nonlinear springs is constructed, and simulations of a TW with the boundary conditions of Eq. (6) are performed using finite element analysis (FEA). Under identical boundary conditions, the numerical predictions based on Eq. (3) and the FEA give similar wave speed and wave shape (see the Supplemental Material [34]).

III. TRANSITION WAVE WITHOUT DISSIPATION

Numerical results for various impact speeds V_{imp} show that the system supports a range of TW speeds $\bar{c}$ [Fig. 2(a)], which is in agreement with previous analysis from the energy perspective. The TW speed increases monotonically with $\bar{V}_{\text{imp}}$ and gradually approaches a plateau. Dispersion analysis (see the Supplemental Material [34]) on the linearized version of Eq. (3) shows that the smallest phase speed of linear waves c_{linear} is approximately $1.2V$ [Fig. 2(a), hatched region], where V is the characteristic speed of the system as defined in Eq. (2). With the boundary conditions of Eq. (6), the upper bound of the TW speed

$\bar{c}$ is around $0.8V$, which is smaller than c_{linear} , indicating that the TW is subsonic [Fig. 2(a) yellow region]. Additionally, for small impact speeds, the response is limited to localized snap-through and deformation near the boundary, rather than a dynamic TW [Fig. 2(a), blue region]. Two representative kinks, with $\bar{c} = 0.4$ and 0.8 , are shown in Figs. 2(e) and 2(f). In the absence of dissipation ($\bar{\beta} = 0$), once initiated, the TW propagates at a constant speed, maintains its shape, and transitions the foundation from state O ($\bar{w} = 0$) to state B ($\bar{w} = 1$). An antikink under the same boundary conditions propagates with the same speed and exhibits identical behavior to a kink, except that the transition direction from B to O is reversed. The relationship between TW speed $\bar{c}$ and the boundary impact speed $\bar{V}_{\text{imp}}$ [Fig. 2(a)] provides guidance for controlling the TW speed in this system.

The upper bound of the TW speed is related to the limited energy input into the system by the boundary impact. Figure 2(b) presents the system's kinetic energy $\bar{E}_k$, elastic energy $\bar{E}_e$, and the total energy $\bar{E}_{\text{tot}} = \bar{E}_k + \bar{E}_e$ at different wave speeds:

$$\begin{aligned} \bar{E}_k &= \frac{1}{2} \int_{-\infty}^{\infty} \left(\frac{\partial \bar{w}}{\partial \bar{t}} \right)^2 d\bar{x}, \\ \bar{E}_e &= \frac{1}{2} \int_{-\infty}^{\infty} \left(\frac{\partial^2 \bar{w}}{\partial \bar{x}^2} \right)^2 + \left(\frac{1}{2} \bar{w}^4 - \bar{w}^3 + \frac{1}{2} \bar{w}^2 \right) d\bar{x}. \end{aligned} \quad (7)$$

All forms of energy above are normalized by the characteristic energy $E = \frac{1}{2} k w_B^4 L^2$. Figure 2(b) shows that all forms of energy increase with wave speed $\bar{c}$. By energy conservation, the total wave energy must equal the energy input from the boundary impact; thus, both the total energy and the resulting wave speed are inherently bounded. As the wave speed approaches zero, the elastic energy remains nonzero. The nonzero elastic energy arises from the static DW. While the relationship between kinetic energy and wave speed is straightforward, the dependence of elastic energy on wave speed results from the dependence of the wavefront shape on the wave speed [Figs. 2(c) and 2(d)].

A notable feature of the TW shape in this system is that in front and after the wavefront center, the system exhibits overshoot behavior [Figs. 2(e)(iii) and 2(f)(iii)] rather than transitioning directly between two stable states. For comparison, the classical Klein-Gordon equation with ϕ^4 potential, often applied to 1D multistable metamaterials [39], has an exact solution of the form $1 + \tanh(\bar{x} - \bar{c}\bar{t}/\sqrt{(1 - \bar{c}^2)})$. In this solution, although the TW also narrows as the wave speed increases, its amplitude (or wave height) remains constant, and no overshoot occurs. In our system, the overshoot arises naturally from the moment balance in the bistable beam-foundation system: without overshoot (i.e., with a smoother tanh-like profile), the distributed forces would balance, but a residual net moment from the bistable foundation would remain. To quantify

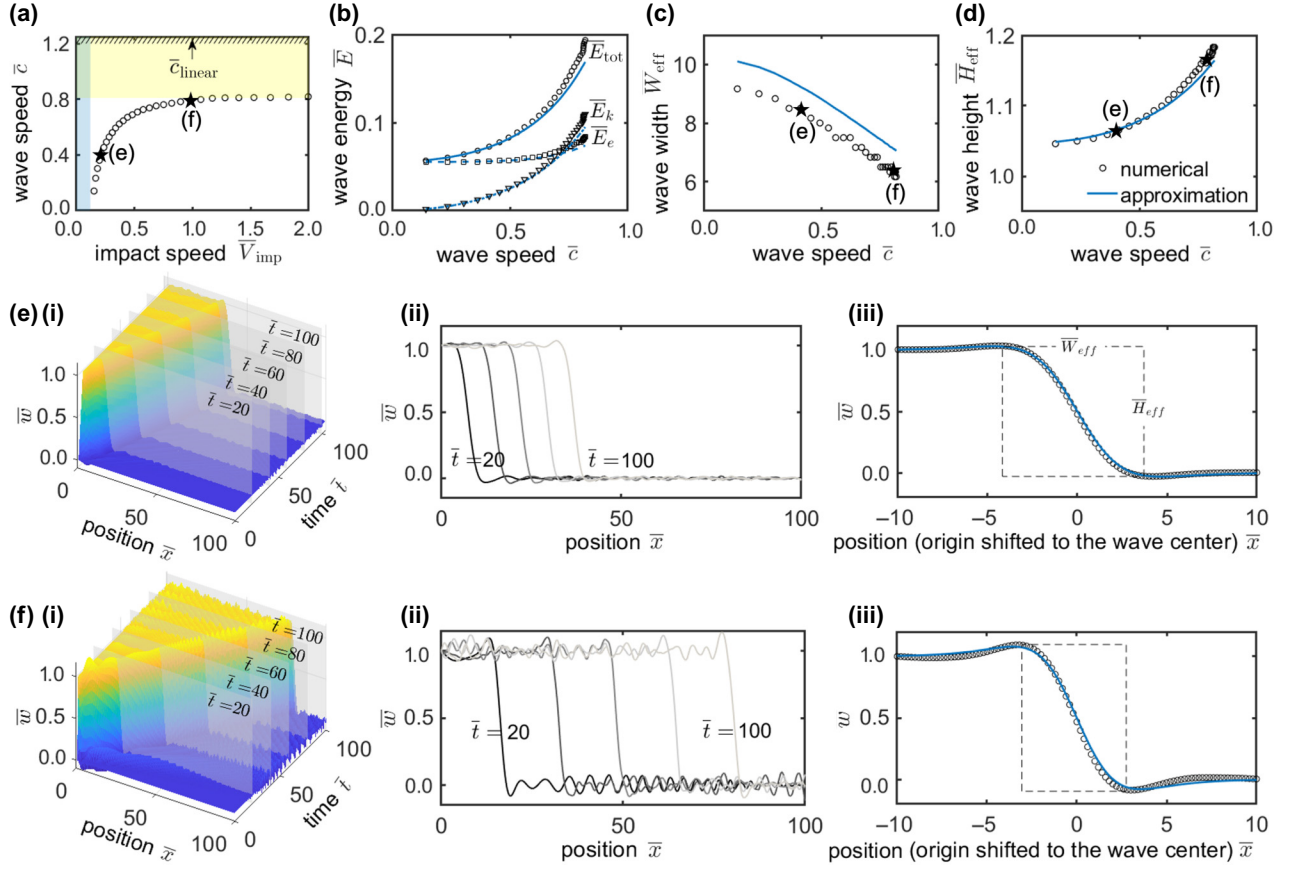


FIG. 2. Numerical results and approximate solutions of the governing ODE Eq. (3) under different impact speeds $\bar{V}_{\text{imp}}$ with $\bar{\beta} = 0$: (a) the relationship between TW speed $\bar{c}$ and impact speed $\bar{V}_{\text{imp}}$; (b) the total wave energy $\bar{E}_{\text{tot}}$, elastic energy $\bar{E}_e$, and kinetic energy $\bar{E}_k$; (c) the relationship between wave speed $\bar{c}$ and effective wave width $\bar{W}_{\text{eff}}$; (d) the relationship between wave speed $\bar{c}$ and wave overshoot amplitude (effective height) $\bar{H}_{\text{eff}}$; (e), (f) are two cases with normalized TW speeds $\bar{c} = 0.4$ and $\bar{c} = 0.8$, respectively, showing (i) the spatiotemporal displacement contour, (ii) the spatial profile at the indicated times, and (iii) the time-averaged spatial profile.

the TW shape, we introduce an effective wave width $\bar{W}_{\text{eff}}$ and height $\bar{H}_{\text{eff}}$, defined by the horizontal and vertical distances between the two overshoot peaks (the maximum and minimum displacements), respectively. As the wave speed decreases, the wave becomes wider [Fig. 2(c)], and the overshooting becomes less pronounced [Fig. 2(d)]. The inclusion of inertia in the dynamic case amplifies the overshoot, analogous to how faster TWs in the Klein-Gordon system yield steeper fronts, and this trend is also consistent with a previous study on similar beam-bistable foundation systems [40]. The left limits of Fig. 2(c) where $\bar{c}$ is zero show that the characteristic width $\bar{W}_{\text{eff}}$ of a static DW is around $9L$, where L is the characteristic length of the system defined in Eq. (2). From Eq. (2) it is obvious that a larger stiffness ratio of the beam to the foundation leads to a wider static DW.

Because of this overshoot, the tanh function does not satisfy Eq. (3). Since analytical solutions are difficult to find (the details on some attempts for the analytical solutions [41–43] are included in the Supplemental

Material [34]), we employ an approximate solution [44] to reflect the relationship between wave energy, shape, and speed. Note that numerical approaches for finding the shape of the transition front exist [29] and could be implemented for this specific system. Subsequent analysis of the stability of this front could also be indicative of its mobility as a function of damping.

The constant wave speed and traveling solitary wave behavior observed from the numerical results (in the absence of dissipation) suggest that an approximate solution takes the form

$$\bar{w}(\bar{x}, \bar{t}) \approx a_0 + \sum_{i=1}^n a_i \tanh^{2i-1} \left(\frac{\bar{\xi}}{\bar{W}} \right), \quad (8)$$

where $\bar{\xi} = \bar{x} - \bar{c}\bar{t}$. Substituting Eq. (8) into Eq. (3) results in a nonzero residual $r(\bar{\xi}, \bar{c}, \bar{W}, a_i)$. Minimizing the integrated squared residual $R(\bar{c}, \bar{W}, a_i) = \int_{-\infty}^{+\infty} r^2 d\bar{\xi}$ with respect to $\bar{W}$ and a_i establishes the quantitative relation between $\bar{c}$, $\bar{W}$, and a_i . The approximate wave energy at

various speeds $\bar{c}$ is calculated using Eqs. (7) and (8). The first-order approximation ($n = 1$) can only capture the qualitative trends between wave shape, energy, and speed but lacks the overshoot characteristic (see the Supplemental Material [34]) since it only contains the tanh function. The second-order approximation ($n = 2$), shown as blue lines in Figs. 2(b)–2(d), 2(e)(iii), and 2(f)(iii), captures the quantitative dependence of wave shape and energy on wave speed better, and can also capture the overshoot properly.

IV. TRANSITION WAVE WITH DISSIPATION

In realistic systems, dissipation is present ($\bar{\beta} > 0$) and can be utilized to stop the TWs. Numerical results show that dissipation causes the TW to decelerate, lose energy, and eventually stop [Fig. 3(a)]. After the TW stops, it forms a stable static DW that separates two adjacent stable regions (stability analysis of the static DW is included in the Supplemental Material [34]). A higher impact speed $\bar{V}_{\text{imp}}$ yields a greater initial wave speed $\bar{c}_{\text{init}}$ and higher wave energy, as discussed in Figs. 2(a) and 2(b). Meanwhile, a smaller dissipation coefficient $\bar{\beta}$ slows the decay of wave energy. These two factors together result in a longer propagation distance $\bar{D}$. It is observed from Fig. 3(a) that $\bar{c}(\bar{t})$, the tangent slope of the phase boundary, approximately follows the form

$$\bar{c}(\bar{t}) = \bar{c}_{\text{init}} e^{-\bar{t}/\tau} \quad (9)$$

where τ is the characteristic decay time. The propagation distance $\bar{D}$ can then be expressed as

$$\bar{D} = \int_0^\infty \bar{c}(\bar{t}) d\bar{t} = \bar{c}_{\text{init}} \tau. \quad (10)$$

To determine τ , the energy balance between dissipation rate and the rate of total energy loss is used:

$$-\frac{\partial \bar{E}_{\text{tot}}}{\partial \bar{t}} = \bar{\beta} \int_{-\infty}^\infty \left(\frac{\partial \bar{w}(\bar{x}, \bar{t})}{\partial \bar{t}} \right)^2 d\bar{x}. \quad (11)$$

Figure 3(a) shows the approximate wave energy distribution at each time $\bar{t}$, calculated by Eqs. (8) and (7) with the instantaneous wave speed $\bar{c}(\bar{t})$, along with the numerical energy distribution. The good agreement between the approximated and numerical results enables a reasonable assumption that during the deceleration, the wave profile depends only on the current wave speed $\bar{c}(\bar{t})$, irrespective of how $\bar{c}(\bar{t})$ changes over time. Hence, Eq. (11) simplifies to

$$-\frac{\partial \bar{E}_{\text{tot}}}{\partial \bar{t}} = \bar{\beta} F(\bar{c}(\bar{t})) \quad (12)$$

where F is a function of $\bar{c}(\bar{t})$. Integrating Eq. (12) over time and using Eq. (9) gives

$$\begin{aligned} \bar{E}_{\text{tot}}(\bar{c}_{\text{init}}) - \bar{E}_{\text{tot}}(\bar{c} = 0) &= \bar{\beta} \tau \int_0^\infty F(\bar{c}_{\text{init}} e^{-\bar{t}/\tau}) d\bar{t} \\ &= \bar{\beta} \tau G(\bar{c}_{\text{init}}) \end{aligned} \quad (13)$$

and thus τ can be expressed as $\tau = \Delta \bar{E}_{\text{tot}}(\bar{c}_{\text{init}}) / \bar{\beta} G(\bar{c}_{\text{init}})$. Finally, substituting this into Eq. (10) yields

$$\bar{\beta} \bar{D} = \bar{c}_{\text{init}} \frac{\Delta \bar{E}_{\text{tot}}(\bar{c}_{\text{init}})}{G(\bar{c}_{\text{init}})}. \quad (14)$$

Equation (14) shows that $\bar{\beta} \bar{D}$ is a function of $\bar{c}_{\text{init}}$. To verify this result, we numerically sweep $\bar{V}_{\text{imp}}$ and $\bar{\beta}$ and plot $\bar{\beta} \bar{D}$ against $\bar{c}_{\text{init}}$ in Fig. 3(b). The overlap of each line confirms the validity of the above analysis. This plot, together with Fig. 2(a), provides practical guidance for achieving target propagation distances $\bar{D}$, simply by controlling the boundary impact speed $\bar{V}_{\text{imp}}$ for a given damping coefficient $\bar{\beta}$. Complex beam shapes can be formed by sequential kinks and antikinks, forming multiple DWs (Fig. 4). To utilize the relationship between propagation distance and boundary impact speed of a single TW to achieve target configurations, the stopping positions of successive

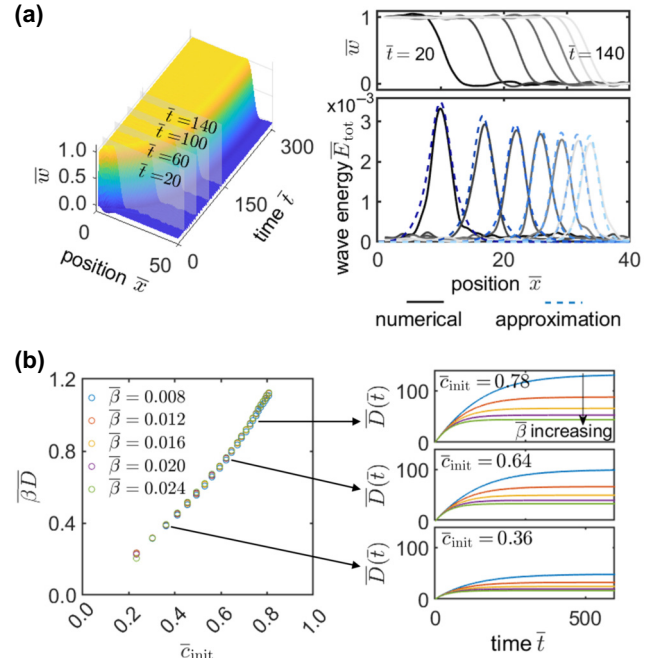


FIG. 3. Numerical results and approximate solutions for partial differential equation (3) with $\bar{\beta} > 0$: (a) The spatiotemporal displacement contour, spatial profile at different times, and total energy distribution at different times, with initial wave speed $\bar{c}_{\text{init}} = 0.5$ and damping parameter $\bar{\beta} = 0.017$. (b) Propagation distance $\bar{D}$ with different initial wave speed $\bar{c}_{\text{init}}$ and damping $\bar{\beta}$.

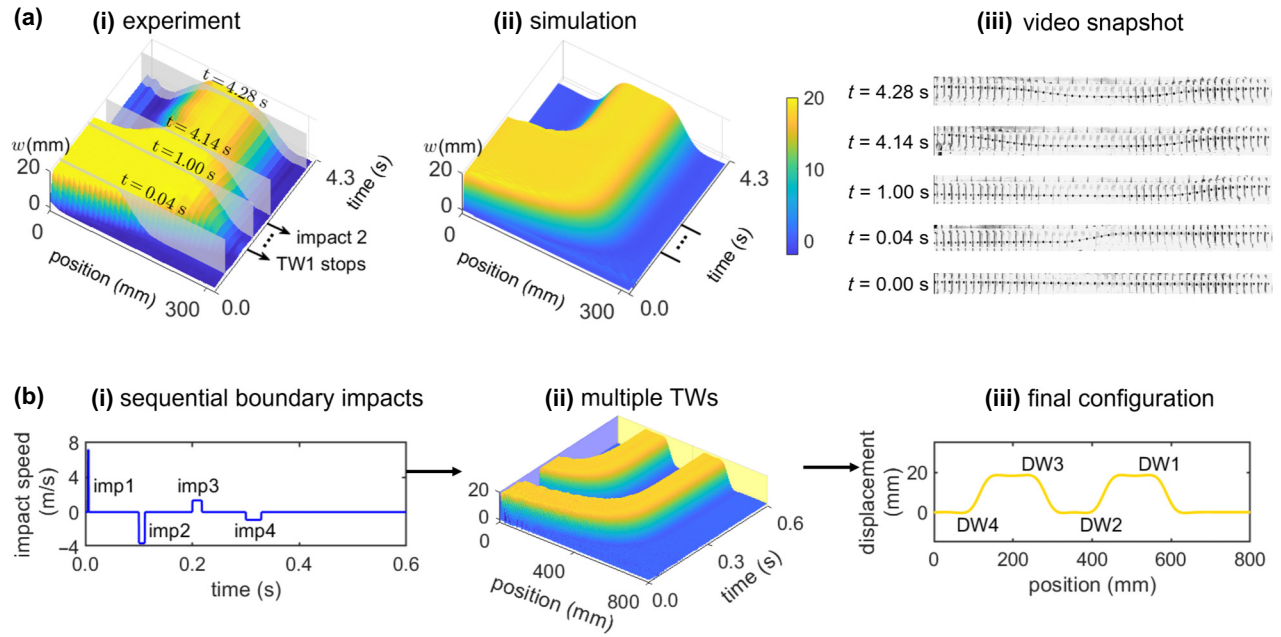


FIG. 4. Multiple TWs and dynamic control of surface shape. (a)(i) Experimental spatiotemporal displacement contour for two TWs, (ii) simulation results using experimental impact speeds obtained by image tracking, and (iii) snapshots of the system configuration at different times by high speed camera. For a better visualization, in (i) and (ii), we compressed the time axis between the moments when TW1 stopped and the onset of the second impact, as indicated by the dotted section on the time axis. (b) Sequential TWs (ii) are triggered by sequential boundary impacts with different impact speeds (i). (iii) Four static domain walls (DWs) are formed after TWs stop, resulting in the final undulated surface shape (configuration).

TWs must be well separated; otherwise, they will interact. (Simulation results and discussion of kink and antikink interactions that occur when they are too close are provided in the Supplemental Material [34].)

V. EXPERIMENTAL VALIDATION

To demonstrate the dynamic properties and the feasibility of the dynamic reconfiguration strategy outlined above, we designed a physical beam-bistable foundation system, as shown in Fig. 5(a). The bistable foundation consists of a 1D array of double-beam (DB) units aligned along the x -direction, with a density of one unit every 9 mm. This density represents a design compromise: it ensures that the foundation is sufficiently dense for the continuous approximation in our model to remain valid, such that the deceleration associated with phonon radiation due to the Peierls-Nabarro effect arising from discreteness [45,46] plays only a minor role (the corresponding analysis on the influence of discreteness on the final propagation distance is provided in the Supplemental Material [34]), and at the same time, the finite discreteness of the foundation is essential for pinning and stabilizing the static DWs at the stopping positions. Each DB unit comprises two identical, initially straight beams connected at the center by a rigid square, with both ends clamped to the top and bottom frames. By decreasing the distance between the frames, a

uniform prestrain is introduced to each DB unit, causing the DB units to buckle. Due to the initial straight configuration, the buckled DB unit exhibits two symmetric stable states (states O and B) with equal elastic energy, which can transition between these states through displacement w . The double-beam design prevents asymmetric buckling modes during snap-through transitions [47,48], which would cause bending-torsion coupling of the whole system. A slender beam is attached to the midpoint of each DB unit and spans the length of the system. This beam is free to bend in the x - w plane, ensuring that only forces in the w -direction are transmitted between the foundation and the beam. Both the DB units and the slender beam are made of polyethylene terephthalate, a material with moderate dissipation to allow the TW to propagate a considerable distance. The system dimensions are carefully designed so that the characteristic length L [as defined in Eq. (2)] is approximately one-quarter of the system's total length. This ensures the ability of the system to present multiple DWs at the same time. To initiate a TW, an impact in the w -direction is applied at one boundary of the system. A high-speed camera positioned above the setup records the motion of the DB units in the x - w plane. The black circular trackers, placed on each DB unit, are visible through the transparent top frame and serve as reference points for image tracking to measure the wave propagation and the dynamic response of the system.

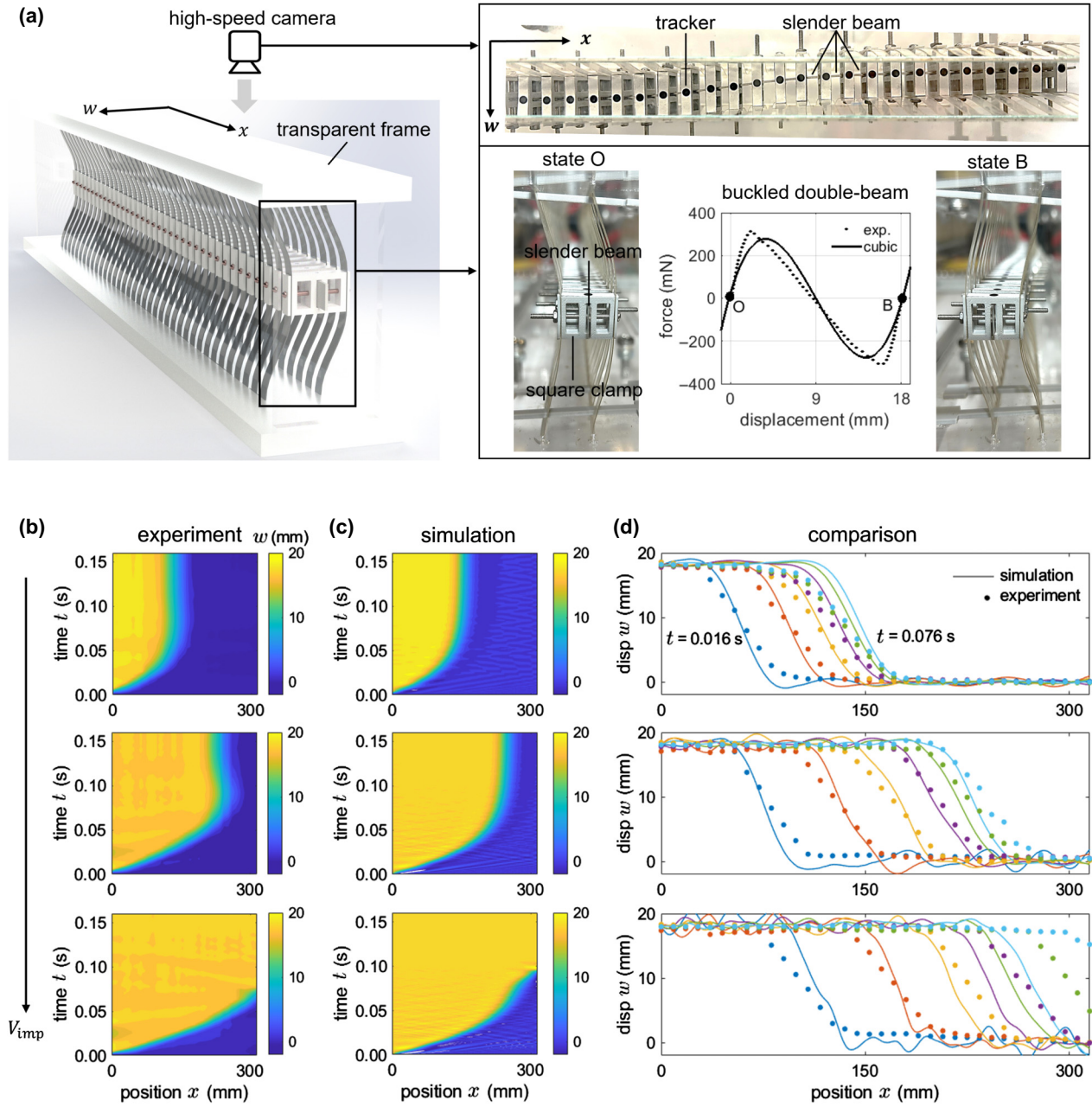


FIG. 5. Experimental setup and comparison between experimental and numerical results. (a) A slender beam coupled to a series of buckled double beam units. (b) Experimental spatiotemporal displacement contour. (c) Numerical spatiotemporal displacement contour. (d) Comparison between numerical and experimental displacement profile at different times.

We first trigger a single TW in the system. Figure 5(b) presents experimental spatiotemporal displacement contours, with increasing boundary impact speeds shown from top to bottom. Figure 5(c) presents the corresponding numerical results, where the experimentally measured displacement time-history of the first unit (see the Supplemental Material [34]) is used as the boundary condition for the simulations. Figure 5(d) compares wave profiles at different times between the experiments and simulations. Parameters in the numerical simulation [Eq. (1)], including

the bistable foundation properties k and w_B , slender beam stiffness EI , and density $A\rho$, were obtained from experimental tests (see the Supplemental Material [34]). To account for the discrete nature of the experimental setup, the bistable foundation was modeled as continuous in the numerical simulation, with an effective foundation stiffness k (force per unit length) applied. The linear damping coefficient β was then used as an empirical fitting parameter, determined by matching the simulated and experimental propagation distances for an intermediate-impact case

(see the Supplemental Material [34]), and this same value was then applied to all other impact speeds without further adjustment to test its validity within the parameters range of interest (different boundary impact speeds). This simple damping coefficient in fact approximates the aggregate effect of all TW deceleration mechanisms, including kinetic energy loss caused by phonon radiation arising from discreteness (P-N effect), material viscosity, and additional frictional losses in the experimental assembly.

As shown in Fig. 5(b), increasing the boundary impact speed significantly increases the wave propagation distance, with the highest-speed case resulting in the wave propagating through the entire system. This experimental observation validates the feasibility of the proposed boundary-impact-induced dynamic reconfiguration method. Despite the simplicity of this model, the simulations in Fig. 5(c) reproduce the experimentally observed trend and the quantitative agreement across all three cases is satisfactory, validating the effectiveness of the model in predicting the key properties of the TWs.

Minor discrepancies are also observed. First, the simulated TW width is slightly smaller than in experiments [Fig. 5(d)], likely because the simulation employs an averaged foundation stiffness per unit length, whereas the experimental stiffness was measured for an individual double-beam unit.

Second, a slight backward rebound of the TW front is observed in the intermediate-impact experiment [Fig. 5(b)], a phenomenon absent in the other tests and in the simulations. To our knowledge, such behavior has not been reported previously and is likely due to (1) small compliance in the clamping frames that maintain prestrain, leading to transient variation in frame spacing and local potential gradient; and (2) local nonuniformities in the bistable potential (e.g., fabrication defects) near two-thirds of the beam length that perturb the propagating TW. To assess the influence of defects, additional simulations (see the Supplemental Material [34]) show that a point defect with stiffness variation less than 20% alters the propagation distance by less than 10% for moderate-speed TWs, whereas larger defects can trap or repel TWs.

Third, small-amplitude oscillations following the boundary impact appear in the simulations but are minor in experiments. We found that the frequency range caused by the BC impact falls within a strongly dissipated frequency band, as identified in frequency-sweep tests (see the Supplemental Material [34]). Because the model employs a simplified linear viscous damping, amplitude- and frequency-dependent dissipation mechanisms in the real system further suppress these oscillations.

Next, sequential TWs were triggered by applying repeated impacts to one boundary. Due to the limited length of the experimental system, only a case with two TWs was tested experimentally [Fig. 4(a)(i)]. The first (kink) TW was initiated from the left boundary with a large

impact speed, resulting in a longer propagation distance. After the first wave stopped, a second (antikink) TW was triggered from the same boundary using a smaller impact speed. The impact speeds were carefully chosen to ensure that the propagation distances of the two waves were well separated, preventing dynamic interactions or overlap of the DW regions. The final configuration of the system, shown in the DIC snapshot in Fig. 4(a)(iii), consists of two DW regions dividing the system into three stable sections. The numerical simulation in Fig. 4(a)(ii), using the same parameters validated for a single TW and the impact speeds from the experimental test with two waves, shows good agreement with the experimental result. To further demonstrate the feasibility of our dynamic reconfiguration strategy, multiple sequential TWs were simulated in a longer system. In Fig. 4(b)(i), four sequential impacts with decreasing impact speeds are applied on the left boundary of the system, resulting in four TWs with different initial wave speeds in Fig. 4(b)(ii) and different stop positions in Fig. 4(b)(iii). After all these TWs stop, four static DW regions form, resulting in the final undulated surface shape. All movies corresponding to the experimental cases are provided in the Supplemental Material [34] (see Movies S1–S4).

VI. CONCLUSION

In summary, we present a comprehensive numerical, analytical, and experimental investigation of TW propagation in a beam coupled to an elastic foundation composed of symmetrical bistable units. This symmetric bistable energy landscape supports two-way TWs (i.e., kinks from state O to state B and antikinks from state B to state O). We show that the system supports TWs with tunable propagation speeds that can be controlled through the boundary impact speed. By incorporating a linear viscous damping model, we investigate the relationships among wave speed, propagation distance, and damping coefficient, from which we establish a strategy for dynamically reconfiguring the beam surface morphology. Sequential boundary impacts with distinct speeds generate alternating kinks and antikinks that arrest at different positions, leading to the coexistence of multiple static DWs and the formation of a stable, globally undulated surface shape.

This boundary-driven dynamic strategy, which controls the propagation distance of TWs (i.e., the positions of the DWs) is a simple and effective approach for controlling the global shape of the system from a single location, which could be of use in scenarios where complex, multipoint control and actuation are not possible. Moreover, because the relationship between the impact speed and the resulting TW properties is normalized by the characteristic length and time of the system, our approach is applicable for controlling surface shape across both macro and micro scales.

Extending this work to a 1D system with a parameter gradient in the symmetric energy landscape, as well as to 2D systems, could broaden the property space and enhance the controllability of reconfiguration, ultimately enabling the realization of more complex surface shapes.

Finally, the current study has several limitations that suggest directions for future work. The precise control of the reconfigured surface shape depends on both the model fidelity, especially the energy dissipation mechanism, which governs transition-wave deceleration and propagation distance, and the accuracy of the boundary excitation that initiates the waves. Further improvement in these aspects will be critical for realizing practical applications. In addition, quantitatively characterizing the role of discreteness in this system will clarify how phonon radiation and DW stability are balanced, thereby enhancing the robustness of the proposed reconfiguration strategy.

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DATA AVAILABILITY

The data supporting this study's findings are available within the article.

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